

NIELIT Virtual Academy

INTERSHIP PROSPECTUS

Name of the Internship: Online Internship in Artificial Intelligence and Machine Learning using Python

Internship Code: IN31

Mode of Conduction: Online

Starting Date: 2nd July 2025

Last date of registration: 30th June 2025

Duration: 6 Weeks

Objective of the Course

- ❖ Provide basic theoretical concepts underlying machine learning.
- ❖ Provide a comprehensive understanding of artificial and machine learning concepts.
- ❖ Provide deep understanding with an effective algorithmic Perspective of AI & ML concepts.
- ❖ Provide hands-on to solve real-life AI & ML problems.

Outcome of the Course

- ❖ Provide the participants with in-depth knowledge AI-ML and its implementation

Course Fee: Rs: 1000/- (inclusive of GST)

Eligibility: Pursuing an Undergraduate level course or above

Methodology

- ✓ Teaching Mode: Self-Pace
- ✓ Access from anywhere anytime
- ✓ Content Access through e-learning portal
- ✓ Doubt Removal Session
- ✓ Covers both Theory & Practical
- ✓ Certification: On completion of the Mini Project

Registration Link: <http://nva.nielit.gov.in>

Contact Details:

- ✓ Course coordinator Name: Ms. Maithreyees S
- ✓ Email: trng.chennai@nielit.gov.in; ds@nielitchennai.edu.in
- ✓ Mobile number: 7598730125

Course Structure:

Module No	Module Title
1	Introduction to Machine Learning & Data pre-processing
2	Machine Learning I
3	Machine Learning II
4	Artificial Neural Network
5	Deep Learning
6	Mini Project

Syllabus:

Detailed Syllabus

Module 1. Introduction to Machine Learning & Data pre-processing

- ❖ Objectives, Applications and types of Machine Learning
- ❖ Python libraries suitable for Machine Learning
- ❖ Dimensionality reduction-Principal Components Analysis
- ❖ Practical Implementation using Python
- ❖ Assignment-1

Module 2. Machine Learning I

- ❖ Building linear regression& KNN classifier
- ❖ Support Vector Machines
- ❖ Decision Tree
- ❖ Practical Implementation using Python
- ❖ Assignment-2

Module 3. Machine Learning II

- ❖ Unsupervised Learning – Introduction, K-Means Algorithm
- ❖ Fuzzy clustering
- ❖ Density Based Clustering
- ❖ Practical Implementation using Python
- ❖ Assignment-3

Module 4. Artificial Neural Network

- ❖ Introduction- Building Neural Network, Activation functions and Loss functions
- ❖ Training a single perceptron
- ❖ Back Propagation Algorithm
- ❖ Practical Implementation using Python
- ❖ Assignment - 4

Module 5. Deep Learning

- ❖ Introduction, CNN with Deep learning
- ❖ RNN- Architecture and Training
- ❖ LSTM and its application
- ❖ Practical Implementation using Python
- ❖ Assignment-5

Module 6. Project

- ❖ Capstone project
- ❖ Project Report Submission